# ISyE 3770 HW6

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## Problem 6.1.6

We observe the sample mean is the average across all data points:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Therefore, we have the following:

```
0.19 + 0.78 + 0.96 + 1.31 + 2.78 + 3.16 + 4.15 + 4.67 + 4.85 + 6.50 + 7.35 + 8.01 + 8.27 + 12.06 + 31.75
```

```
## [1] 272.82
```

$$\bar{x} = \frac{272.82}{19} = 14.359$$

We observe the sample standard distribution is defined as the following:

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$

We will use R to speed up the process…

```
breakdown_times <- c(0.19, 0.78, 0.96, 1.31, 2.78, 3.16, 4.15, 4.67, 4.85, 6.50,
                     7.35, 8.01, 8.27, 12.06, 31.75, 32.52, 33.91, 36.71, 72.89)

variance <- sum((breakdown_times - 14.359)^2) / 18
sqrt(variance)
```

```
## [1] 18.88045
```

## Problem 6.2.4

Let's start with the stem-and-leaf plot:

```
octane_ratings <- c(
  88.5, 98.8, 89.6, 92.2, 92.7, 88.4, 87.5, 90.9,
  94.7, 88.3, 90.4, 83.4, 87.9, 92.6, 87.8, 89.9,
  84.3, 90.4, 91.6, 91.0, 93.0, 93.7, 88.3, 91.8,
  90.1, 91.2, 90.7, 88.2, 94.4, 96.5, 89.2, 89.7,
  89.0, 90.6, 88.6, 88.5, 90.4, 84.3, 92.3, 92.2,
  89.8, 92.2, 88.3, 93.3, 91.2, 93.2, 88.9,
  91.6, 87.7, 94.2, 87.4, 86.7, 88.6, 89.8,
  90.3, 91.1, 85.3, 91.1, 94.2, 88.7, 92.7,
```

```r
  90.0, 86.7, 90.1, 90.5, 90.8, 92.7, 93.3,
  91.5, 93.4, 89.3, 100.3, 90.1, 89.3, 86.7,
  89.9, 96.1, 91.1, 87.6, 91.8, 91.0, 91.0
)

stem(sort(octane_ratings), scale = 2)
```

```
##
##   The decimal point is at the |
##
##    83 | 4
##    84 | 33
##    85 | 3
##    86 | 777
##    87 | 456789
##    88 | 23334556679
##    89 | 0233678899
##    90 | 0111344456789
##    91 | 0001112256688
##    92 | 22236777
##    93 | 023347
##    94 | 2247
##    95 |
##    96 | 15
##    97 |
##    98 | 8
##    99 |
##   100 | 3
```

Now, we can calculate the median and quartiles:

```r
median_octane <- median(octane_ratings)
quartiles <- quantile(octane_ratings, type=6)

median_octane
```

```
## [1] 90.4
```

```r
quartiles
```

```
##      0%     25%     50%     75%    100%
##  83.400  88.575  90.400  92.200 100.300
```

## Problem 6.3.2

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

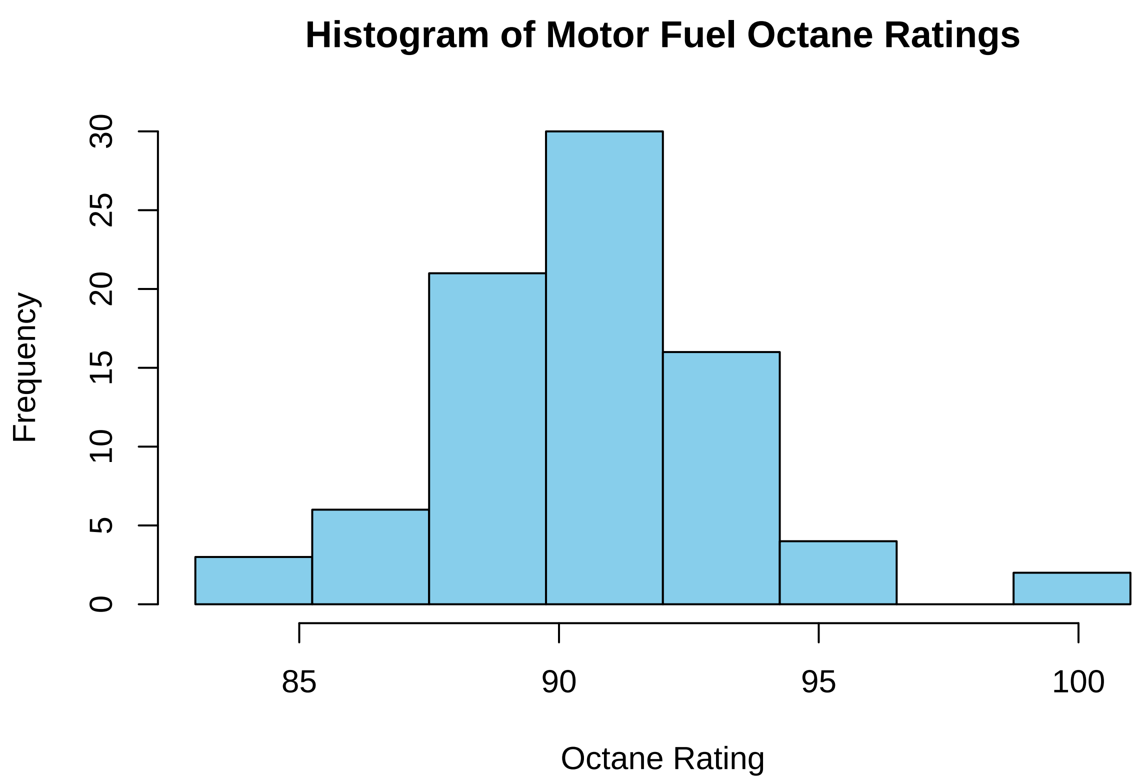
```r
library(knitr)

octane_ratings <- c(
  88.5, 98.8, 89.6, 92.2, 92.7, 88.4, 87.5, 90.9,
  94.7, 88.3, 90.4, 83.4, 87.9, 92.6, 87.8, 89.9,
  84.3, 90.4, 91.6, 91.0, 93.0, 93.7, 88.3, 91.8,
  90.1, 91.2, 90.7, 88.2, 94.4, 96.5, 89.2, 89.7,
  89.0, 90.6, 88.6, 88.5, 90.4, 84.3, 92.3, 92.2,
  89.8, 92.2, 88.3, 93.3, 91.2, 93.2, 88.9,
  91.6, 87.7, 94.2, 87.4, 86.7, 88.6, 89.8,
  90.3, 91.1, 85.3, 91.1, 94.2, 88.7, 92.7,
  90.0, 86.7, 90.1, 90.5, 90.8, 92.7, 93.3,
  91.5, 93.4, 89.3, 100.3, 90.1, 89.3, 86.7,
  89.9, 96.1, 91.1, 87.6, 91.8, 91.0, 91.0
)

num_bins <- 8
breaks <- seq(floor(min(octane_ratings)), ceiling(max(octane_ratings)), length.out = num_bins + 1)
hist_result <- hist(octane_ratings, breaks = breaks, plot = FALSE)
freq <- hist_result$counts

bins <- paste0(head(breaks, -1), " \\( \\leq x < \\) ", tail(breaks, -1))

bins[length(bins)] <- paste0(breaks[num_bins], " \\( \\leq x \\leq \\) ", breaks[num_bins + 1])

relative_freq <- freq / sum(freq)
cumulative_relative_freq <- cumsum(relative_freq)

freq_table <- data.frame(
  Class = bins,
  Frequency = freq,
  `Relative Frequency` = round(relative_freq, 4),
  `Cumulative Relative Frequency` = round(cumulative_relative_freq, 4)
)

kable(freq_table, caption = "Frequency Distribution of Motor Fuel Octane Ratings")
```

Table 1: Frequency Distribution of Motor Fuel Octane Ratings

| Class | Frequency | Relative.Frequency | Cumulative.Relative.Frequency |
|---|---:|---:|---:|
| $83 \leq x < 85.25$ | 3 | 0.0366 | 0.0366 |
| $85.25 \leq x < 87.5$ | 6 | 0.0732 | 0.1098 |
| $87.5 \leq x < 89.75$ | 21 | 0.2561 | 0.3659 |
| $89.75 \leq x < 92$ | 30 | 0.3659 | 0.7317 |
| $92 \leq x < 94.25$ | 16 | 0.1951 | 0.9268 |
| $94.25 \leq x < 96.5$ | 4 | 0.0488 | 0.9756 |
| $96.5 \leq x < 98.75$ | 0 | 0.0000 | 0.9756 |
| $98.75 \leq x \leq 101$ | 2 | 0.0244 | 1.0000 |

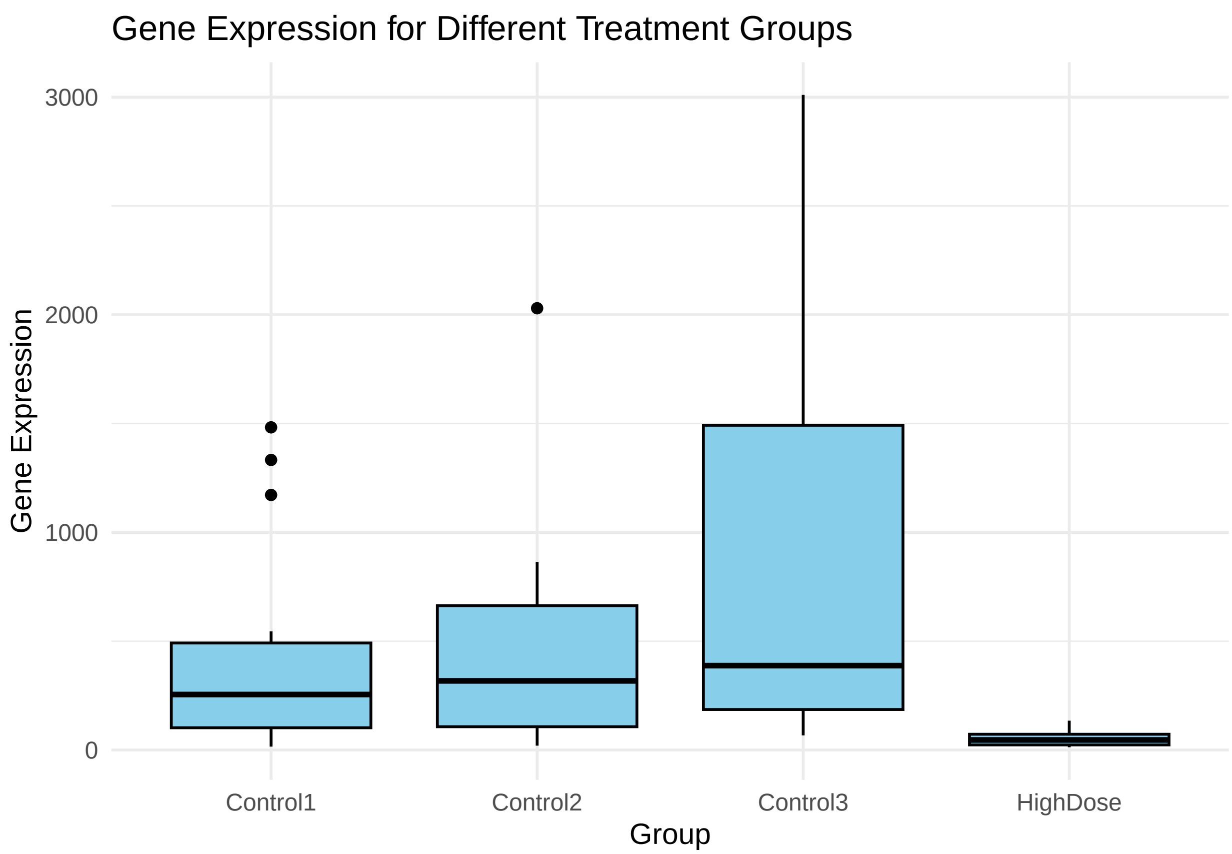
```r
hist(octane_ratings, breaks = breaks, col = "skyblue", border = "black",
     main = "Histogram of Motor Fuel Octane Ratings",
     xlab = "Octane Rating",
```

```
    ylab = "Frequency")
```




**Problem 6.4.9**

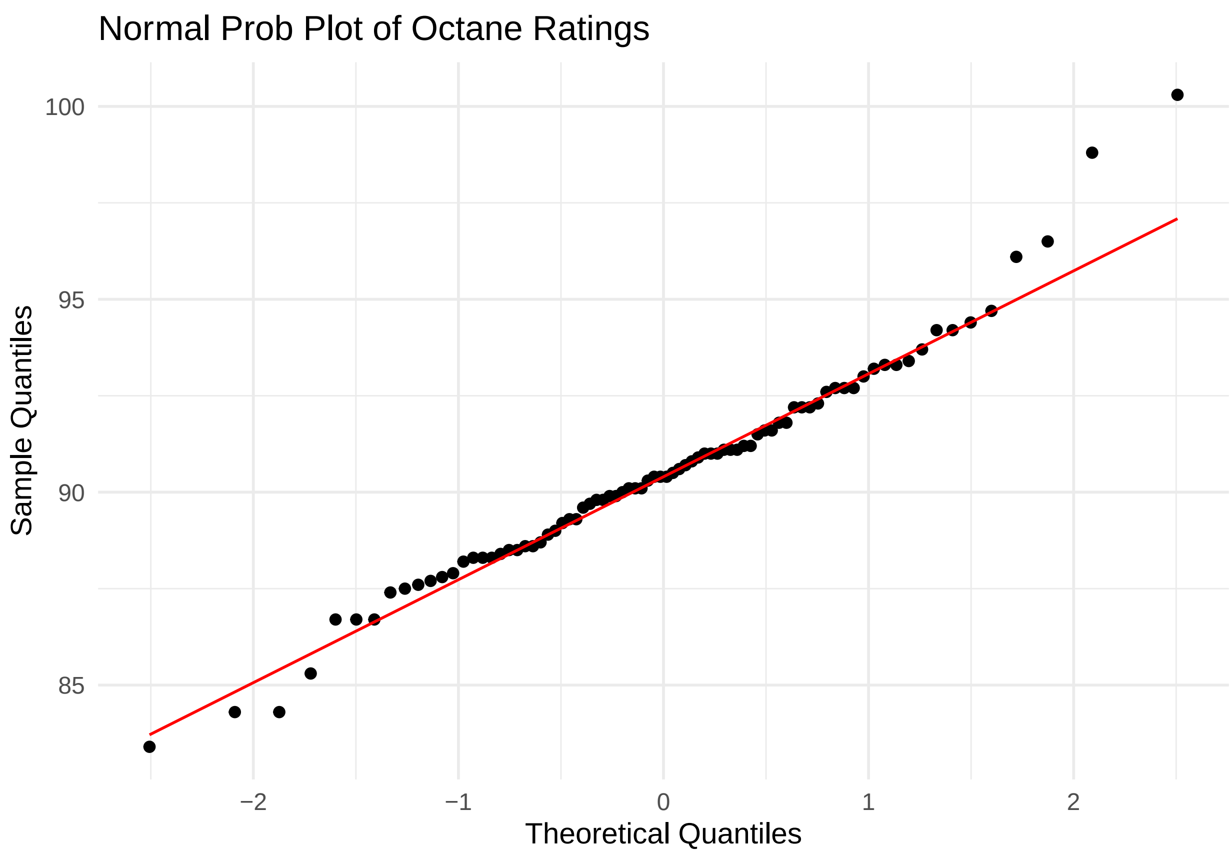
```
library(ggplot2)

data <- read.table("C:/Users/devin/OneDrive/Documents/Georgia_Tech/6-81.txt", header = TRUE)

ggplot(data, aes(x = Group, y = Expression)) +
  geom_boxplot(fill = "skyblue", color = "black") +
  labs(title = "Gene Expression for Different Treatment Groups",
       x = "Group",
       y = "Gene Expression") +
  theme_minimal()
```




The HighDose displays lower gene expression levels than the control groups. The control groups demonstrate greater variability in gene expression; Control3 has multiple outliers which skew the data.

### Problem 6.7.2

```
library(ggplot2)

ggplot(data.frame(octane_ratings), aes(sample = octane_ratings)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  labs(title = "Normal Prob Plot of Octane Ratings",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()
```




Points which lie approximately on the red line indicate a normal distribution.

Points which deviate from the red line suggest deviations from a normal distribution.

Since most of the points lie approximately on the red line, we can conclude the octane rating is normally distributed.